Developer Technology Trends & Job Market Analysis

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12 October 2026



OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



Key findings from the developer survey and job-market data

- Web stacks lead current use: HTML/CSS + JavaScript + TypeScript is the most common language combination (291 respondents)
- Next-year interest favors C#-based web stacks (230) and Python, which appears on its own (104)
- PostgreSQL is the top database today (994) and the most wanted next year (1,265), ahead of Microsoft SQL Server
- AWS leads cloud platforms in both use and interest, followed by Microsoft Azure
- Spring Boot, React and ASP.NET Core are the largest web-framework groups
- Respondents are early-to-mid career: about 69% are aged 25-44 and about 46% hold a bachelor's degree
- Job postings peak in Washington DC (5,316); Swift ($130,801) and Python ($114,383) pay the highest average salaries

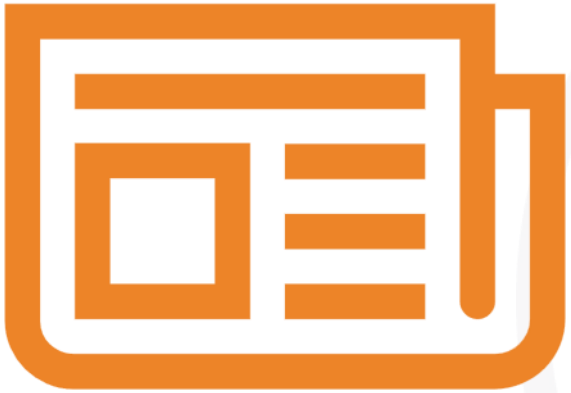
INTRODUCTION



Purpose of the report, target audience and value of the analysis

- Purpose: identify the languages, databases, cloud platforms and web frameworks developers use today, and compare them with what they want to use next
- Target audience: technology leaders, hiring and workforce-planning teams, and training providers
- Value: evidence-based decisions on which skills to hire for, train in and invest in
- Data: developer survey responses, job-posting counts and average language salaries

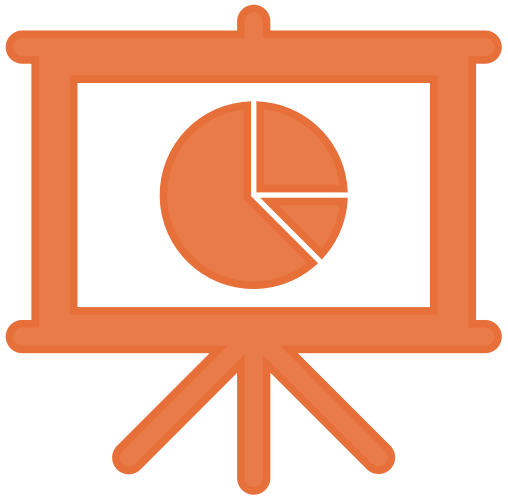
METHODOLOGY



Data sources, data collection and data wrangling steps

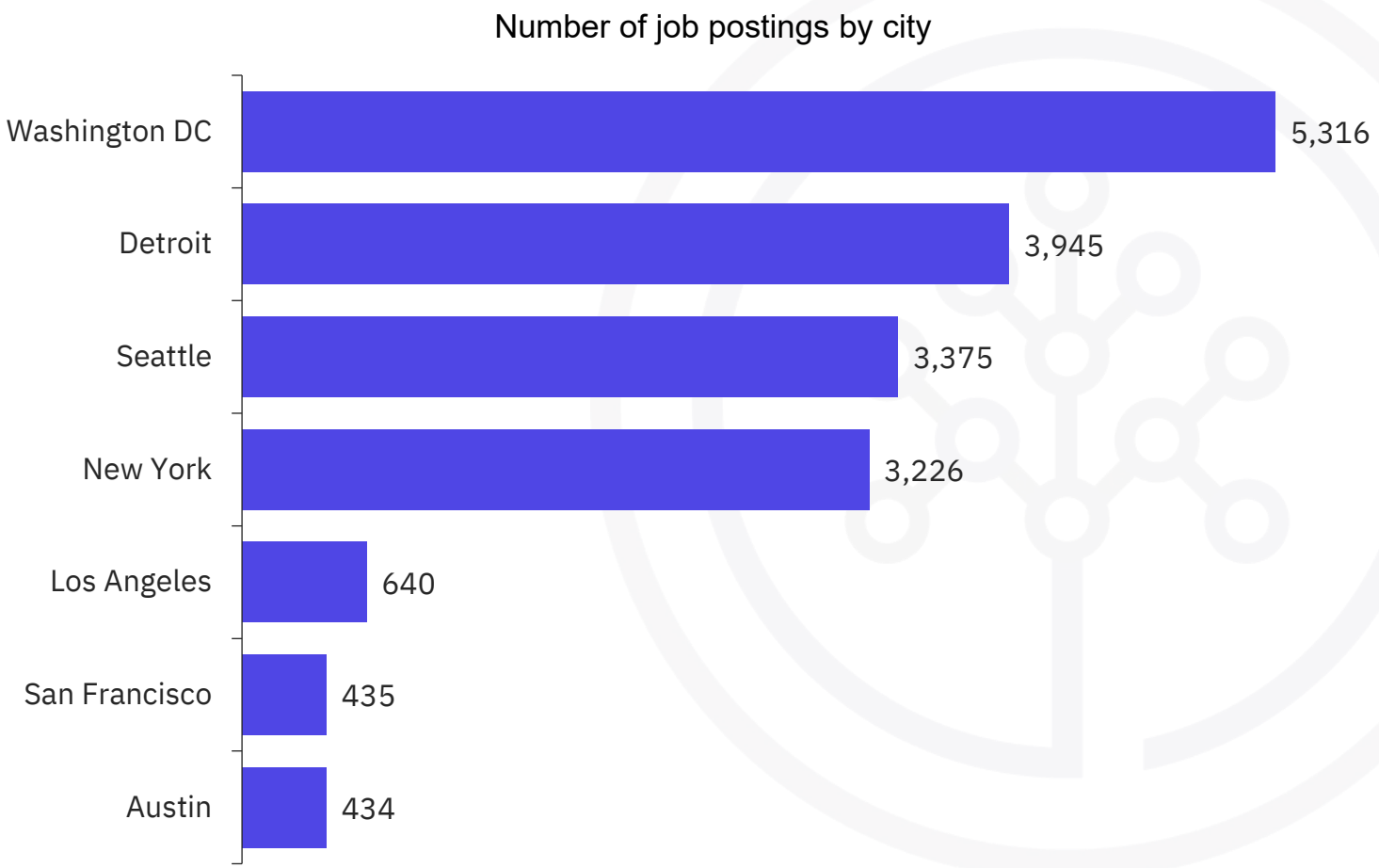
- Data sources: developer survey dataset (survey_data_updated.csv), job-postings.xlsx and popular-languages.csv from IBM / Coursera labs
- Data collection: Job API requests, web scraping of language salaries, and CSV download of the survey data
- Data wrangling: data discovery (checking structure, columns and missing values), then structuring into analysis-ready tables
- Analysis: top-10 counts and a three-tab dashboard in IBM Cognos Analytics; job and salary charts built from the xlsx and csv files

Results



Job market, language trends and database trends

JOB POSTINGS

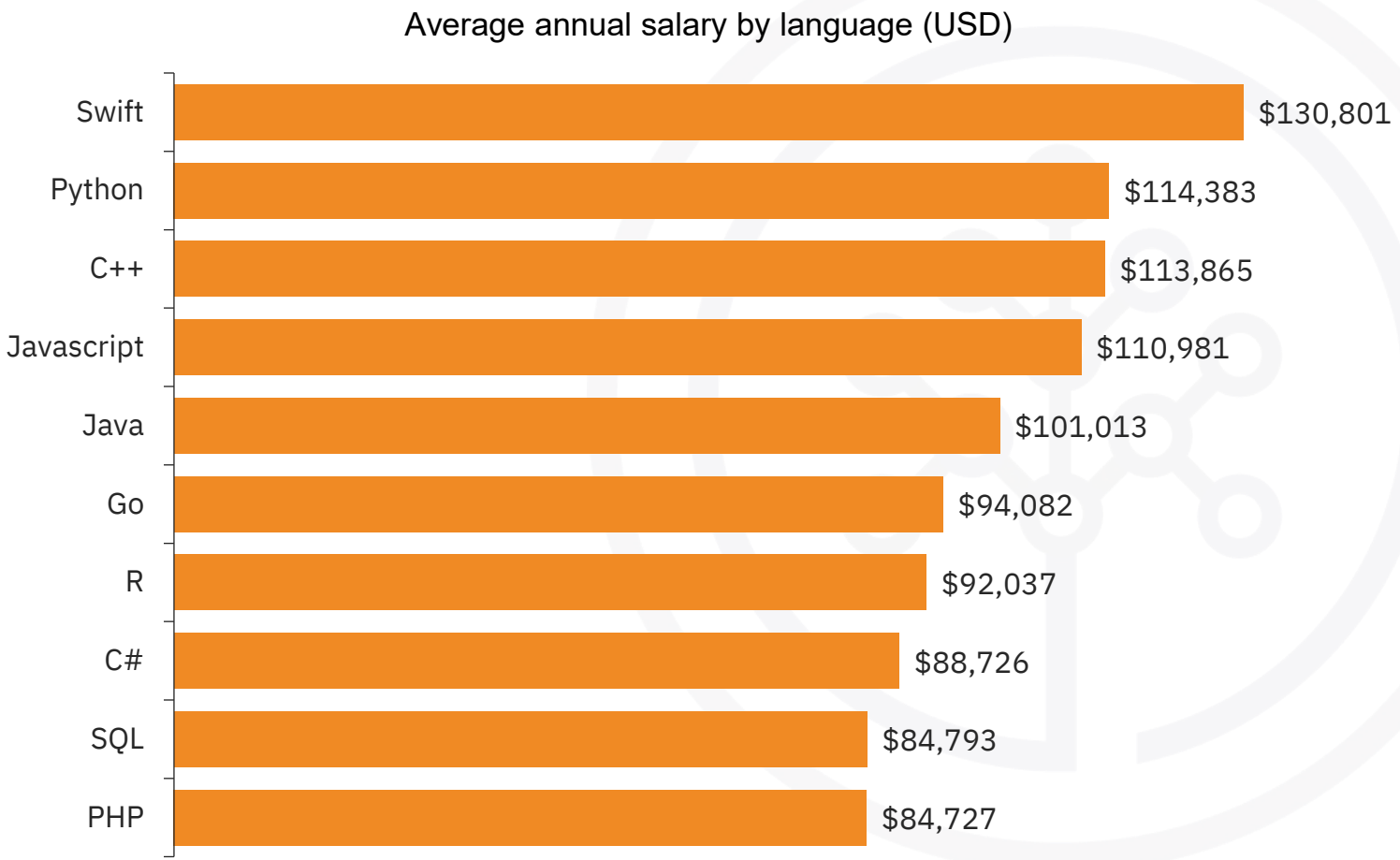


5,316 postings

Washington DC has the most postings, followed by Detroit (3,945), Seattle (3,375) and New York (3,226).

Los Angeles, San Francisco and Austin each have fewer than 700.

POPULAR LANGUAGES



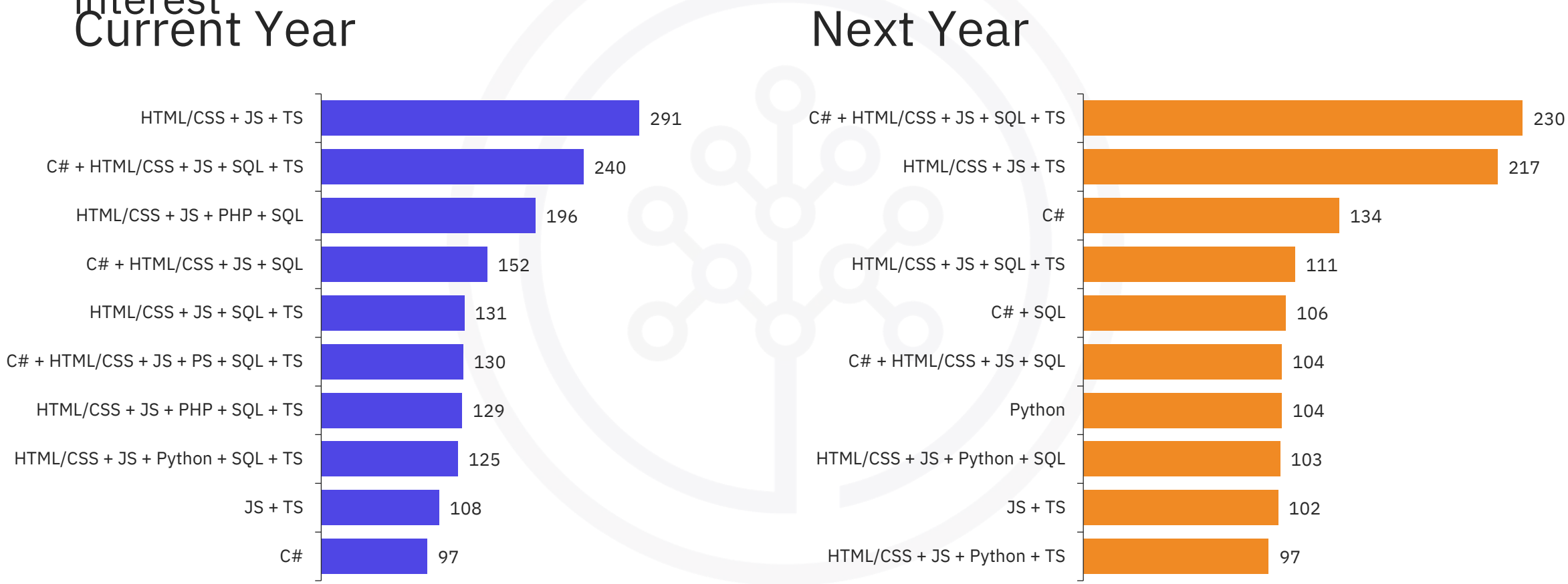
Swift: $130,801

Python ($114,383) and C++ ($113,865) follow, with JavaScript close behind ($110,981).

SQL and PHP are lowest, at about $84.7K.

PROGRAMMING LANGUAGE TRENDS

Web stacks dominate today; C# and Python gain ground in next-year interest



Bars show the top 20 respondent language stacks (combinations). JS = JavaScript, TS = TypeScript, PS = PowerShell.

LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- HTML/CSS + JavaScript + TypeScript is the most common stack today (291), ahead of C#-based web stacks (240)
- Next-year interest shifts toward C#: C# + web + SQL + TypeScript leads (230) and C# alone ranks third (134)
- Python appears in 3 next-year entries (104 on its own) versus 1 stack today

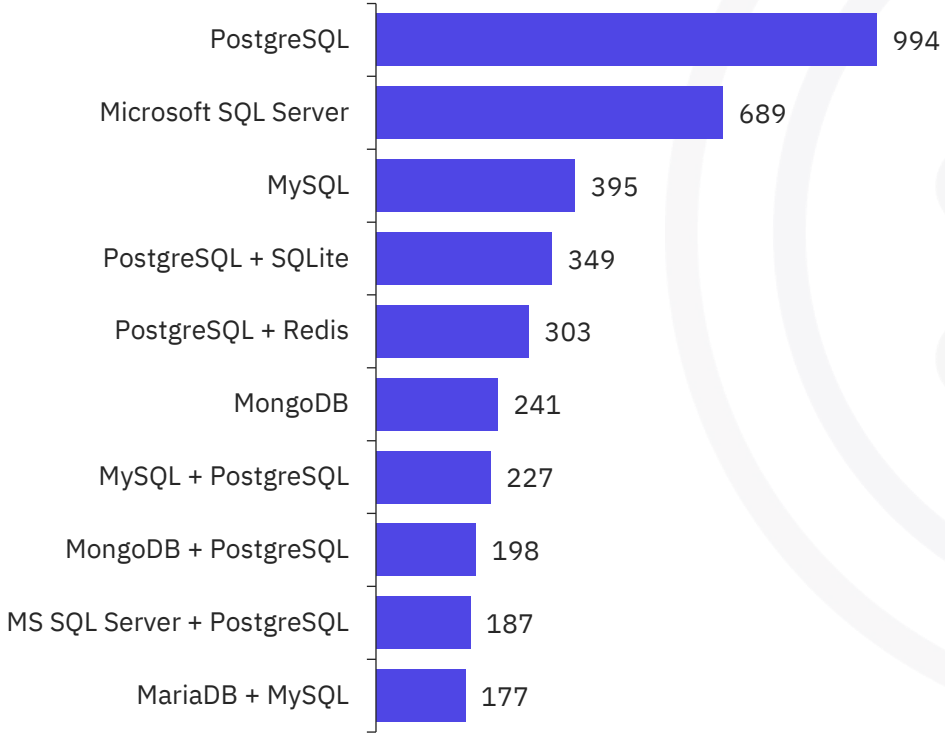
Implications

- Keep web development (HTML/CSS, JavaScript, TypeScript) as the core hiring and training focus
- Add C# and Python learning paths to match growing interest
- Teach languages as stacks: SQL and TypeScript are most often paired with the main web languages

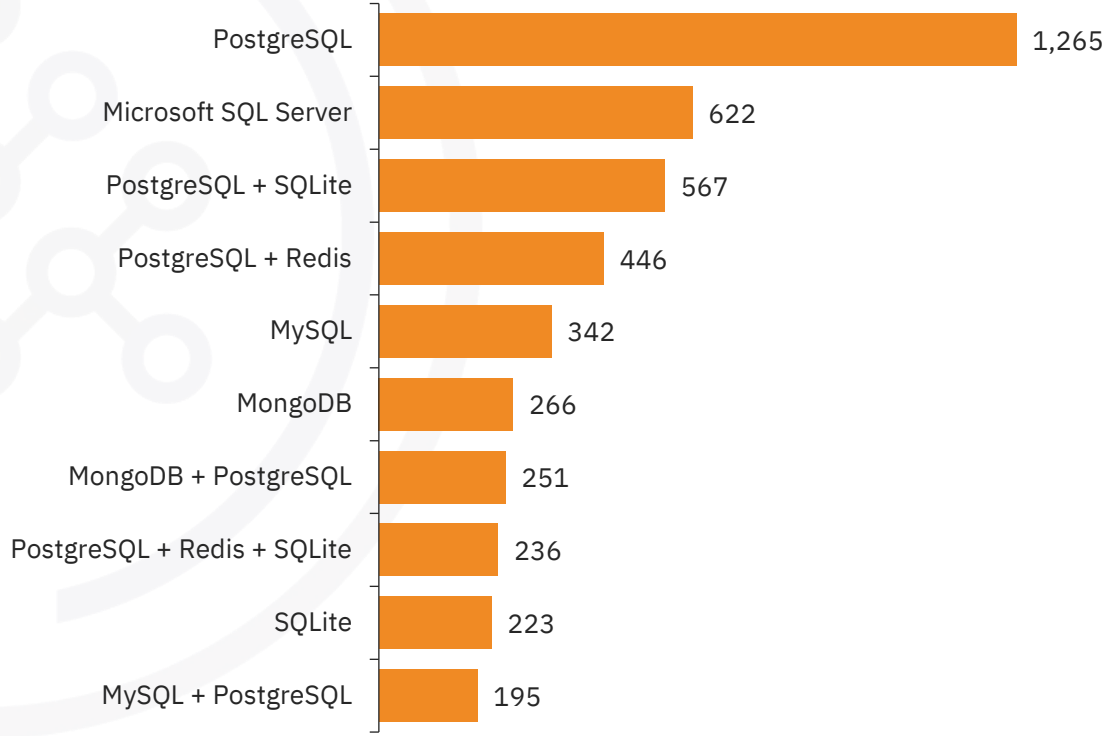
DATABASE TRENDS

PostgreSQL leads both current use and future interest; Microsoft SQL Server is the main alternative

Current Year



Next Year



Bars show the top 10 respondent database combinations; "+" marks databases used together.

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL ranks first today (994) and next year (1,265), about 27% higher
- Microsoft SQL Server is second today (689), but interest dips to 622
- Combined stacks are common: PostgreSQL pairs with SQLite (567), Redis (446) and MongoDB (251)

Implications

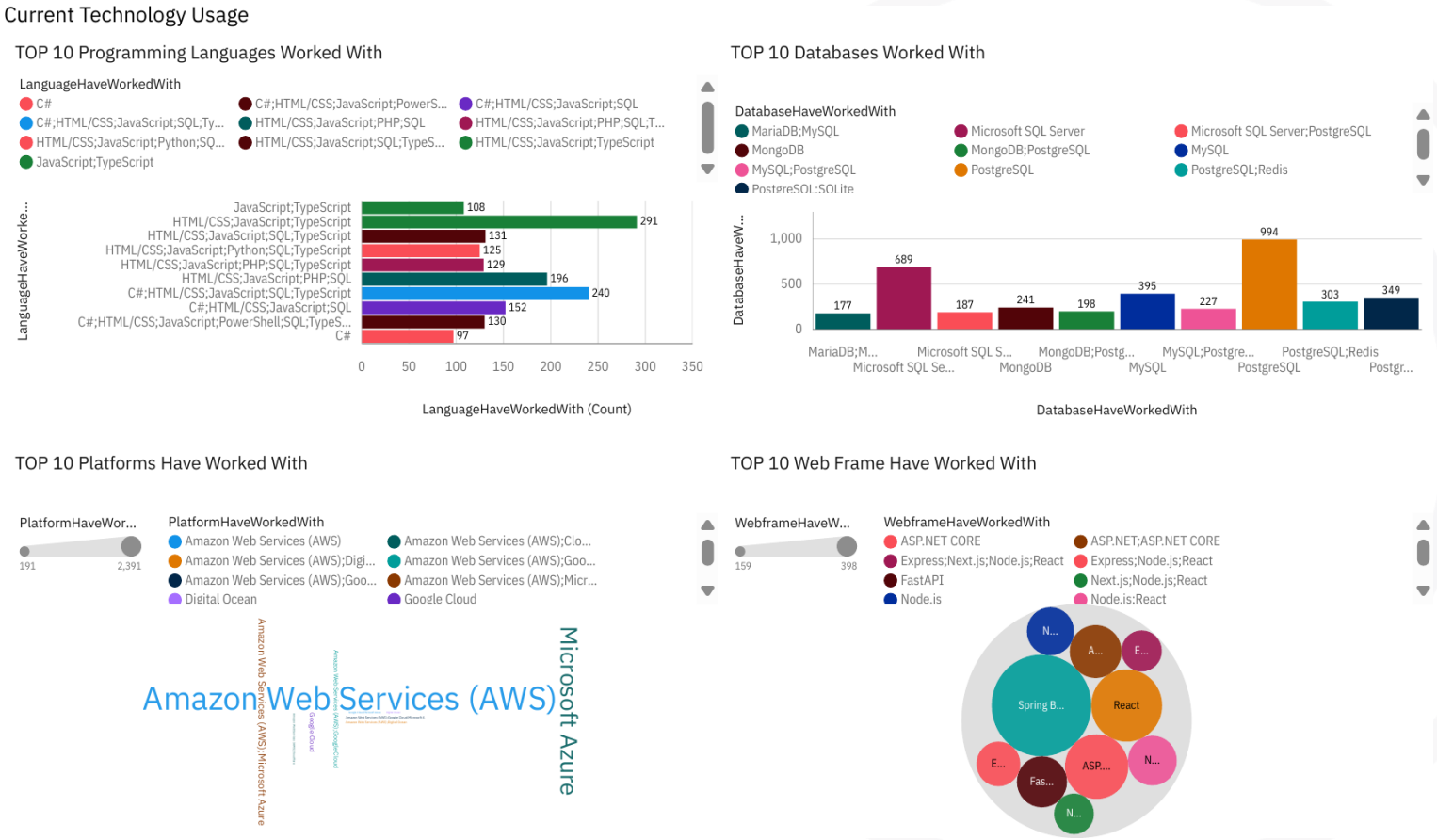
- Prioritize PostgreSQL skills in hiring and training
- Build multi-database skills, e.g. PostgreSQL with Redis for caching or SQLite for lightweight storage
- Keep SQL Server and MySQL support for existing systems while demand shifts toward PostgreSQL

DASHBOARD



Three tabs: current usage, future trends and demographics

DASHBOARD TAB 1: Current Technology Usage



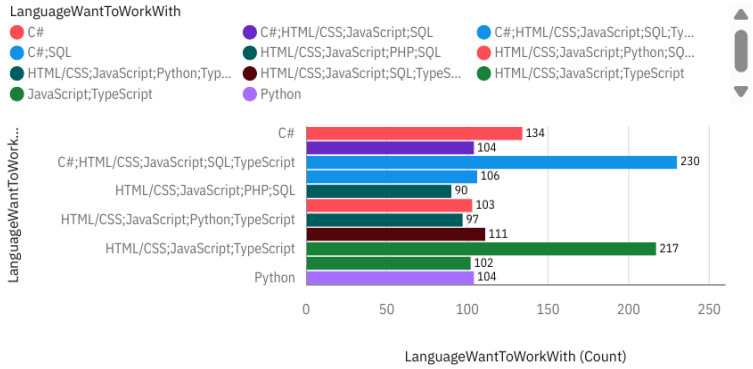
Dashboard Insight

- The most common stack is HTML/CSS + JavaScript + TypeScript (291), then a C# web stack (240)
- PostgreSQL is the most-used database (994), ahead of Microsoft SQL Server (689) and MySQL (395)
- AWS leads cloud platforms, followed by Azure; Spring Boot and React are the largest framework groups

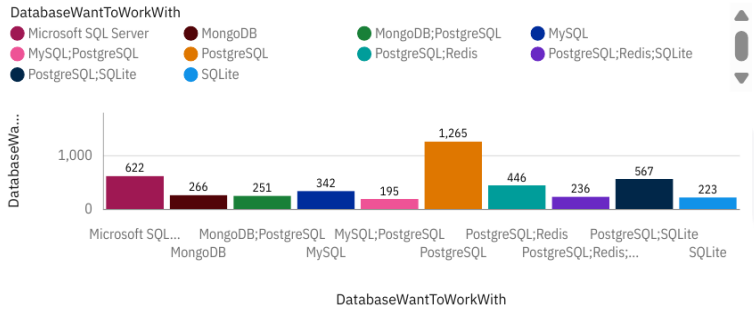
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trend

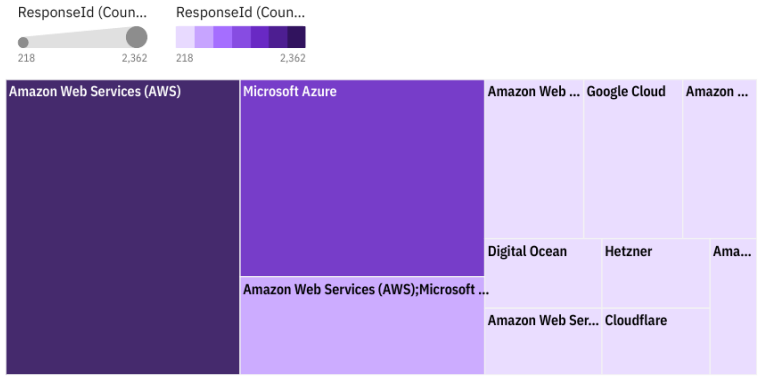
TOP 10 Languages Want to Work With



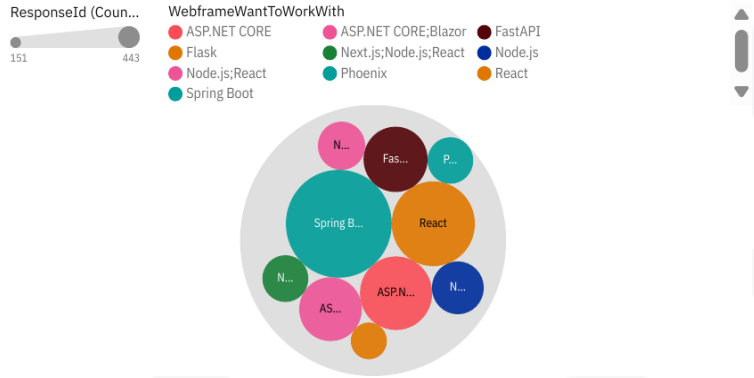
TOP 10 Databases Want to Work With



TOP 10 Platform Want To Work With



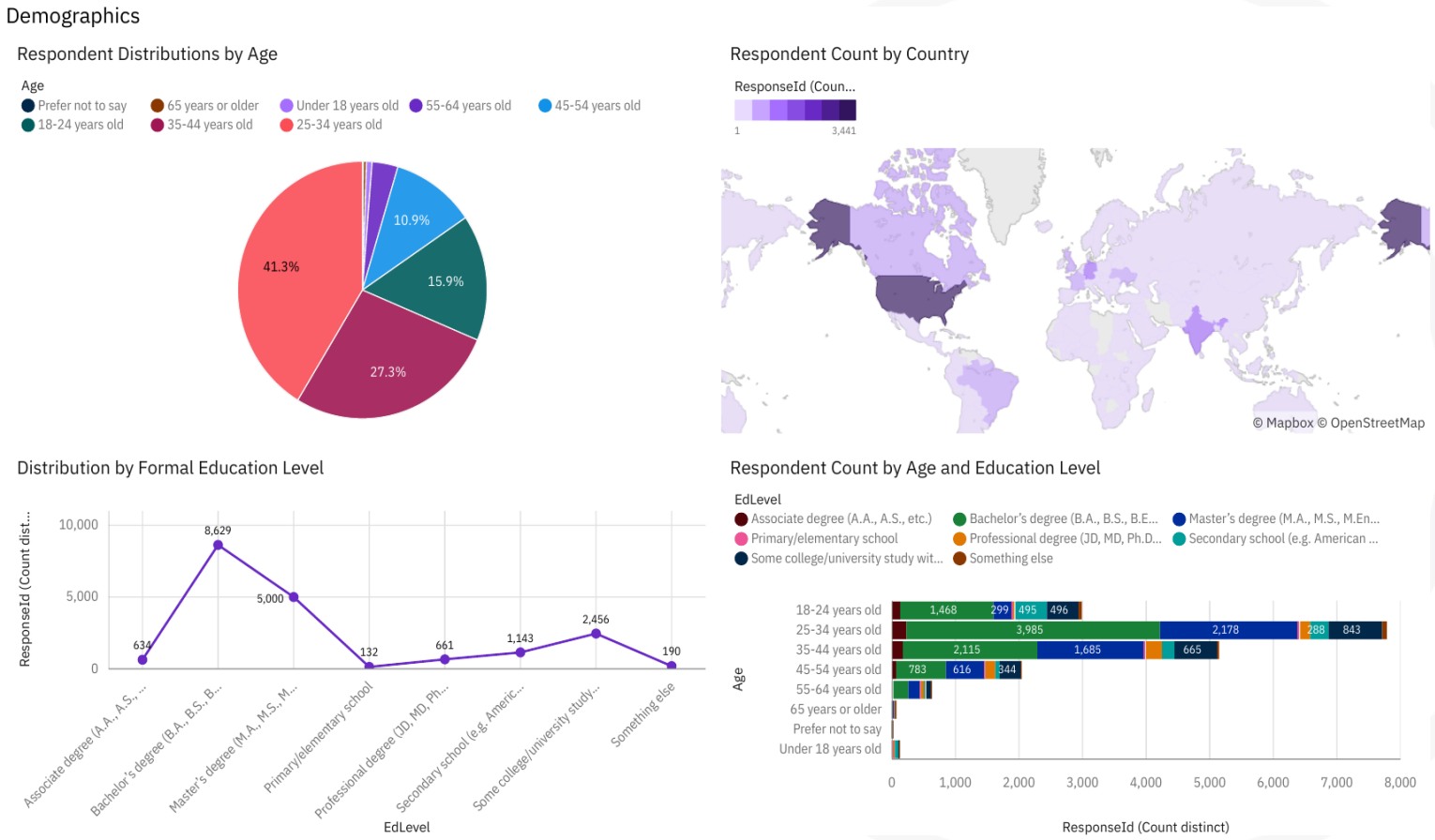
TOP 10 Webframe Want to Work With



Dashboard Insight

- PostgreSQL demand rises to 1,265, about 27% above current use (994)
- Microsoft SQL Server interest dips (622 want vs 689 use)
- Web stacks stay on top (230, 217) and Python appears on its own (104)
- AWS and Azure still lead platform interest

DASHBOARD TAB 3: Demographics



Dashboard Insight

- 25-34 is the largest age group (41.3%); ages 25-44 total about 69%
- Bachelor's degrees are most common (8,629, about 46%), then Master's (5,000, about 27%)
- The US has the most respondents, followed by India
- In the 25-34 group, bachelor's (3,985) and master's (2,178) dominate

DISCUSSION



- Web technologies lead: web stacks top both current use and future interest; Spring Boot and React are the largest framework groups
- PostgreSQL is the clearest database trend, from 994 users to 1,265 who want it
- AWS leads cloud platforms, with Azure second in both use and interest
- Future preferences extend today's stacks (C#, Python) rather than

IBM

OVERALL FINDINGS & IMPLICATIONS

Findings

- Modern web development dominates current use and next-year interest
- PostgreSQL and AWS lead their categories now and in future preference
- Developers work in combined stacks, and the community is mostly aged 25-44 with degrees

Implications

- Keep web development (JavaScript/TypeScript, React, Node.js) as the core hiring and training focus
- Invest in cloud and database skills together, starting with AWS and PostgreSQL
- Develop flexible full-stack, multi-database talent and recruit where demand is highest (Washington DC, Detroit)

CONCLUSION



- Web stack leads: HTML/CSS + JavaScript + TypeScript is the most-used language combination; C# and Python gain interest
- PostgreSQL and AWS on top: PostgreSQL is #1 now and next year, and AWS leads cloud platforms
- Frameworks: Spring Boot, React and ASP.NET Core are the largest groups
- Demand and pay: job postings peak in Washington DC; Swift, Python and C++ lead average salaries

APPENDIX



Respondents by formal education level (share of the eight listed levels)

Education level (dashboard Tab 3)	Respondents	Share
Bachelor's degree	8,629	45.8%
Master's degree	5,000	26.5%
Some college / university	2,456	13.0%
Secondary school	1,143	6.1%
Professional degree	661	3.5%
Associate degree	634	3.4%
Something else	190	1.0%
Primary / elementary school	132	0.7%